

Vend-a-Shu

Complete App Walkthrough

A detailed, step-by-step operating guide for the Vend-a-Shu automated shoe storage system: connecting the app, managing users, photographing shoes with automatic background removal, storing, vending, and returning pairs.

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SECTION 1

System Overview & Requirements

Vend-a-Shu (VAS) is an automated shoe storage system. Each pair of shoes lives in an assigned bin inside the unit. The mobile app is your remote control: it keeps a catalog of every pair (with a studio-quality photo), tells the unit which bin to eject when you want a pair, and guides shoes back to their correct bin when you return them.

How the pieces fit together

- The app (phone or tablet) — the interface you interact with.
- The VAS server & database — keeps the master record of users, shoes, bins, and photos. Runs on the Raspberry Pi inside your unit.
- The storage unit hardware — physical bins organized by column (C1–C6), row (R1–R4), and location (LF/RF). The app addresses bins by these coordinates; when you vend, the correct bin ejects and its LED lights up.

What you need before starting

- A phone or tablet with the Vend-a-Shu app installed.
- Your device on the same Wi-Fi network as the VAS unit.
- A user profile (created in Options) so shoes are stored under the correct owner.
- Camera or photo library access if you want automatic shoe photos — the app will request permission the first time; you can decline and add shoes without photos.

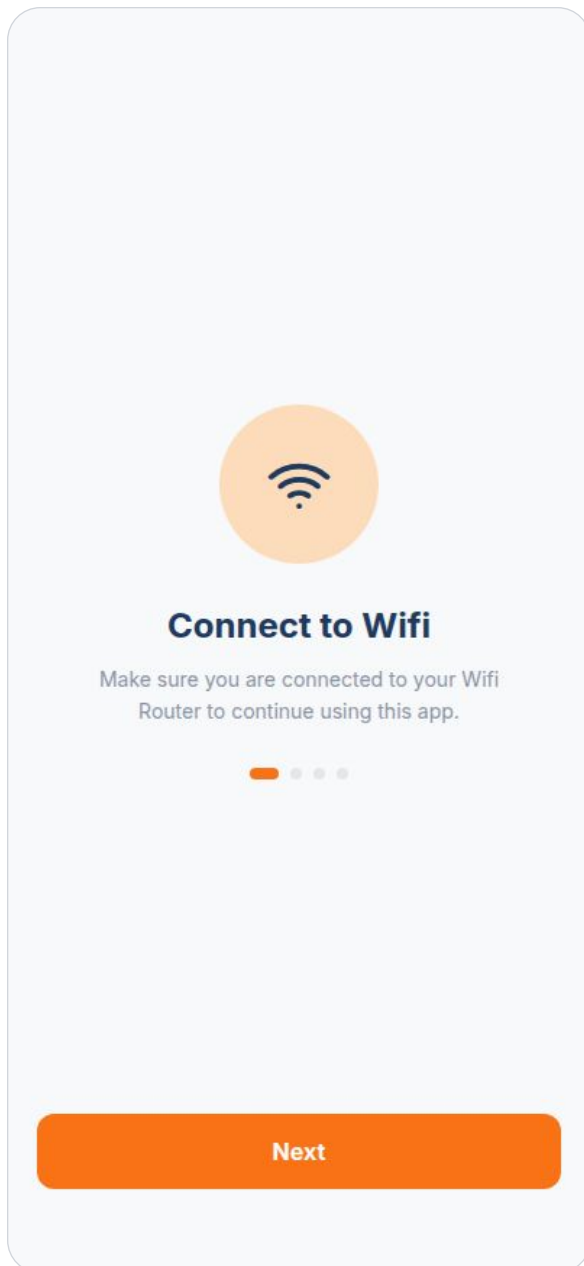
Access your collection from anywhere

- Your VAS data is not locked to your home network: the catalog — every pair, photo, and status — can be accessed remotely from your phone.
- This is especially handy while shopping: pull up your collection in the store to check whether you already own a similar pair, what colors you have, or which designers dominate your closet before buying.
- Vending and returning still require you to be at the unit (the hardware has to physically eject a bin), but browsing, searching, and reviewing your catalog work from anywhere.

Key rules the system enforces for you

- A pair can only move through valid states: stored, then vended, then stored again (or removed). The app will never let a pair be vended twice or returned when it is not out.
- Photos are validated (JPEG/PNG, max 3 MB after processing) and stored with the shoe record, so the catalog thumbnail always matches the pair in the bin.

Onboarding Tour



The very first time you open the app, a short guided tour walks you through what the system does before you touch any controls. Nothing is saved or changed during onboarding — it is purely informational, and you will only see it once.

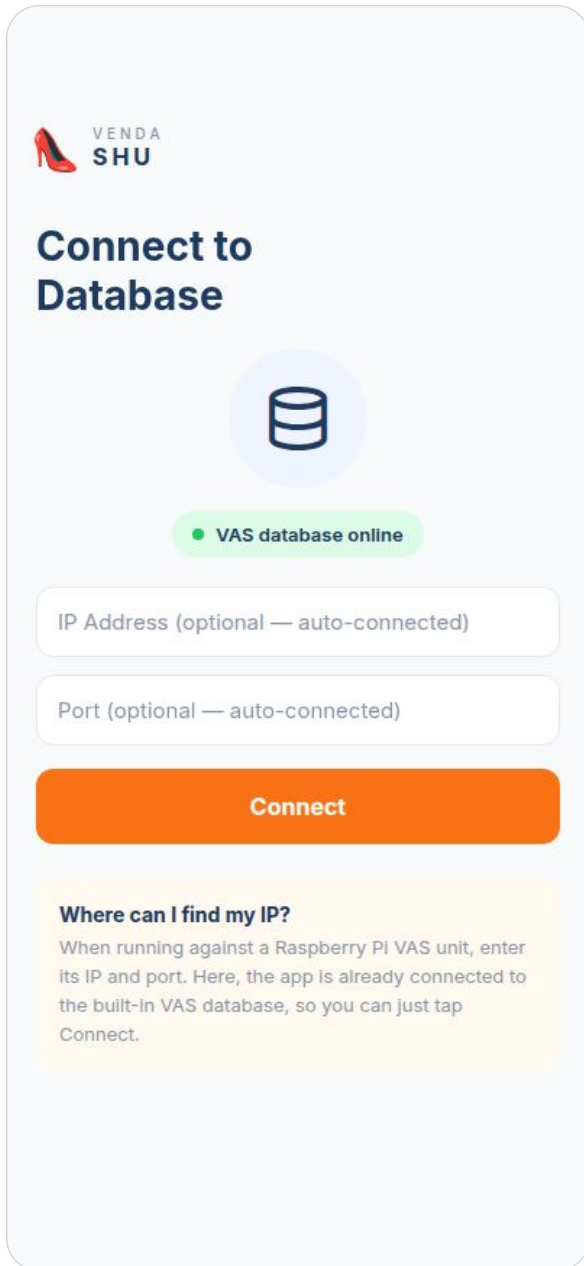
What to do on this screen

- Read the prompt: your device must be on the same Wi-Fi network as the VAS unit. The app communicates with the unit over your local network.
- The dots below the text show your position in the tour (screen 1 of 4). Each screen introduces one concept.
- Tap the orange Next button to advance. There is no data entry anywhere in the tour.

Note for the meticulous:

If you switch Wi-Fi networks later (e.g., a new router), the app does not need to be reinstalled — you will simply re-enter the unit's address on the Connect screen.

Connect to Database



VENDA SHU

Connect to Database



VAS database online

IP Address (optional — auto-connected)

Port (optional — auto-connected)

Connect

Where can I find my IP?
When running against a Raspberry Pi VAS unit, enter its IP and port. Here, the app is already connected to the built-in VAS database, so you can just tap Connect.

This screen establishes the link between the app and the VAS database — the record of every user, shoe, bin, and photo. The app checks connectivity for you: the green “VAS database online” badge means the built-in database has already been found.

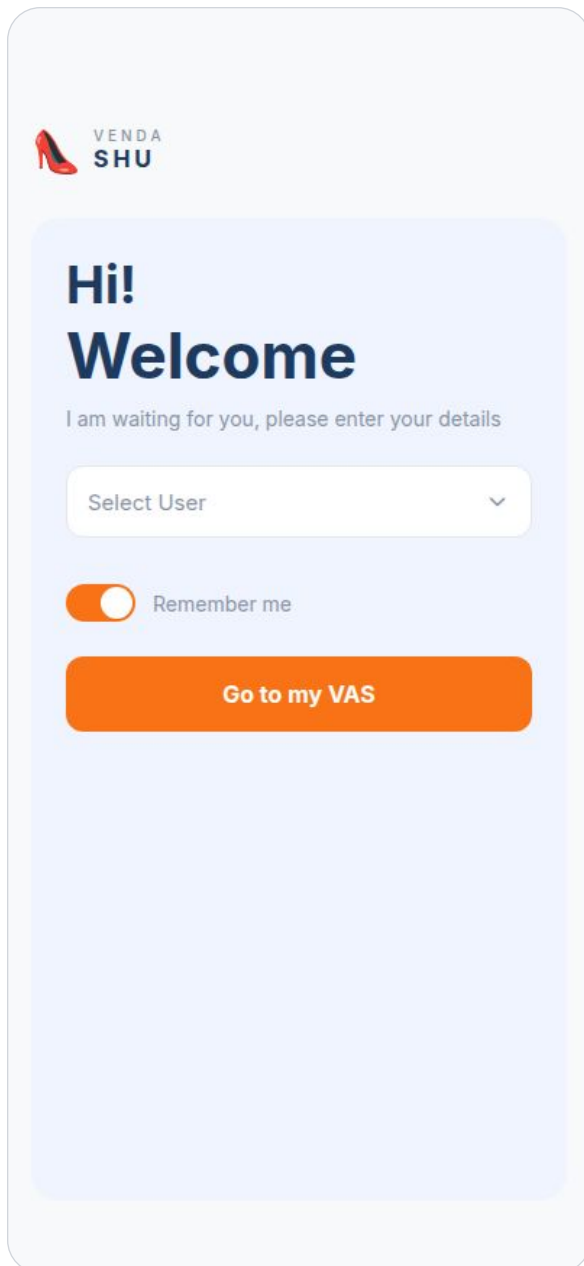
Field-by-field

- IP Address (optional) — your VAS unit's address on your local network, if it is not detected automatically. Find it on the unit's info panel or your router's device list.
- Port (optional) — the unit's network port, supplied with your unit's documentation. Leave blank when the unit is auto-detected.
- Connect — with the green badge showing, just tap Connect. If you entered an IP/port, the app validates the connection before proceeding and tells you explicitly if it fails.
- The yellow help box (“Where can I find my IP?”) is always visible on this screen for reference.

Note for the meticulous:

The connection is remembered for future launches. You only return to this screen if the unit becomes unreachable or you deliberately reconnect to a different unit.

Select Your User



Every shoe in the system belongs to a specific user, so the app needs to know who is operating it. This screen sets the active user for the session.

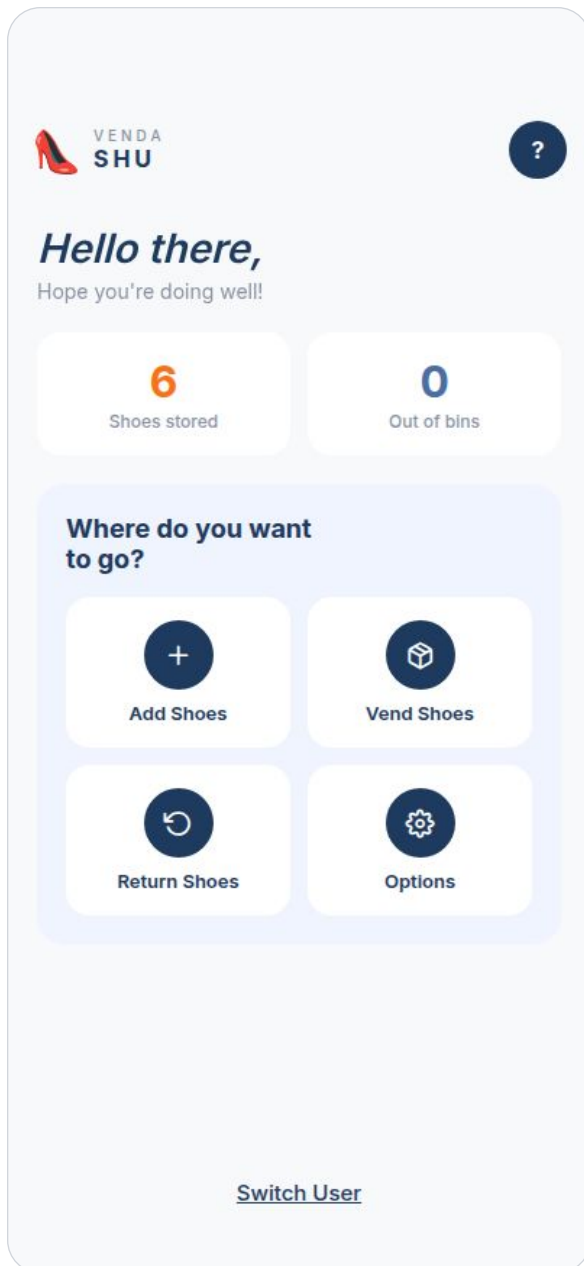
What to do on this screen

- **Select User** — open the dropdown and choose your name. The list comes live from the database; if your name is missing, a user must first be created in Options.
- **Remember me** — leave this on (default) and the app will skip straight to Home on future launches. Turn it off on a shared device if each person should pick their own name every time.
- **Go to my VAS** — confirms your selection and opens the Home screen. The button will not proceed until a user is selected.

Note for the meticulous:

Your selection controls which shoes you see when vending: the catalog can be filtered to the active user's shoes, and new pairs you add are recorded under your name.

The Home Screen

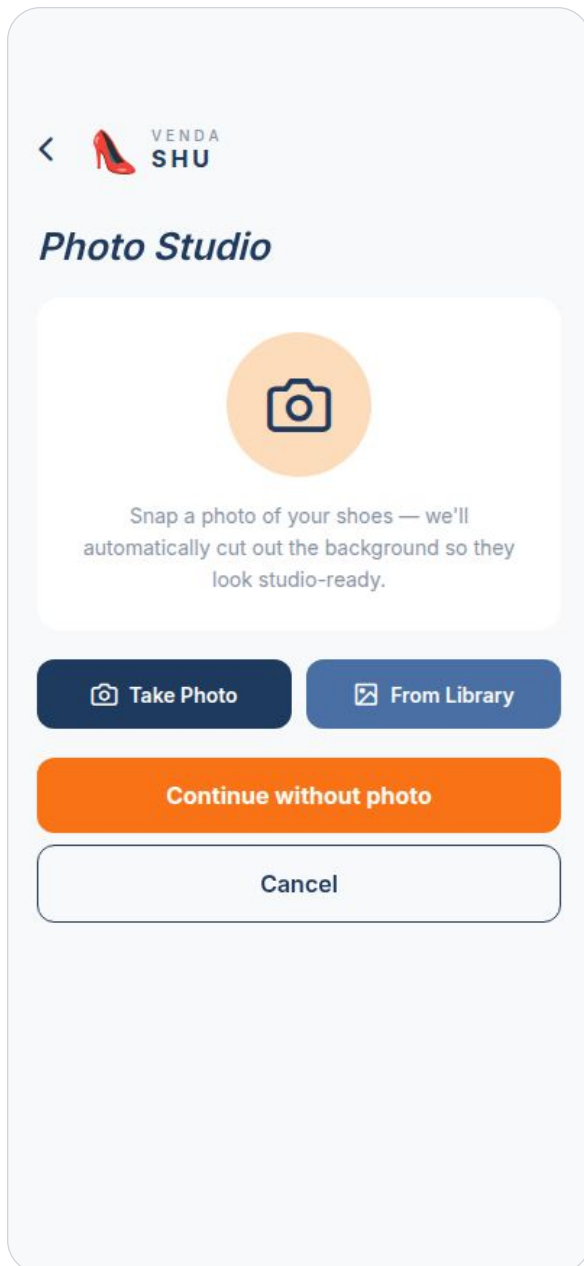


The Home screen is mission control. The two counters at the top are live statistics pulled from the database every time you land here — they are not cached.

Every element, explained

- Shoes stored (orange number) — total pairs currently inside the unit under the active user.
- Out of bins (navy number) — pairs that have been vended and not yet returned. If this is not zero, something is out of the unit right now.
- Add Shoes — begins the two-step intake flow: Photo Studio first, then shoe details and automatic bin assignment.
- Vend Shoes — browse the catalog and have the unit eject a pair.
- Return Shoes — put a vended pair back into its assigned bin.
- Options — user management and system settings.
- Switch User — returns to the user-selection screen without disconnecting from the unit.
- ? (top right) — in-app help.

The Photo Studio



Adding a pair begins with an optional photo. The Photo Studio turns an ordinary phone snapshot into a catalog-quality product image: the app automatically detects your shoes and removes the entire background — no cropping, tracing, or editing on your part.

The full photo workflow

- **Take Photo** — opens the camera (the app asks permission on first use). Best results: shoes on any surface, reasonable lighting, whole pair in frame.
- **From Library** — pick an existing photo instead.
- **Processing** — after you choose an image, a spinner appears for roughly five seconds while the background is removed on the server. The original photo never leaves your VAS system — no third-party photo service is involved.
- **Preview & backdrop** — the cutout appears composited on a clean white background. Tap any swatch in the “Choose a background” row to instantly re-composite onto that backdrop (see next page for the full gallery).
- **Continue without photo** — fully supported. The pair will show a shoe-type icon in the catalog instead of a photo.
- **Cancel** — abandons intake entirely; nothing is saved.

Note for the meticulous:

If processing fails (e.g., an unreadable image), the app tells you exactly why and lets you retry with another photo. Photos are downscaled to 900 px and capped at 3 MB before storage, so intake never bloats the database.

SECTION 7 — ADDING SHOES (2 OF 2)

Backdrop Gallery

Nine backdrop options are available for every photo. “White” is the default catalog look; the eight premade scenes below are professionally generated studio backdrops stored on the VAS server. Switching backdrops re-composites the same cutout — you never need to retake the photo. Your chosen backdrop is baked into the saved image.



Studio Grey

Neutral seamless grey; the classic product-photo look.



Warm Wood

Natural wooden floor with warm tones; homey and organic.



Marble

Polished white marble; bright, premium, high-contrast.



Sunset Orange

Bold orange gradient matching the app's accent color.



Concrete Loft

Smooth polished concrete; modern industrial feel.



Blush Pink

Soft pastel pink studio paper; gentle and fashionable.



Midnight Navy

Deep navy with a soft spotlight; moody and premium.

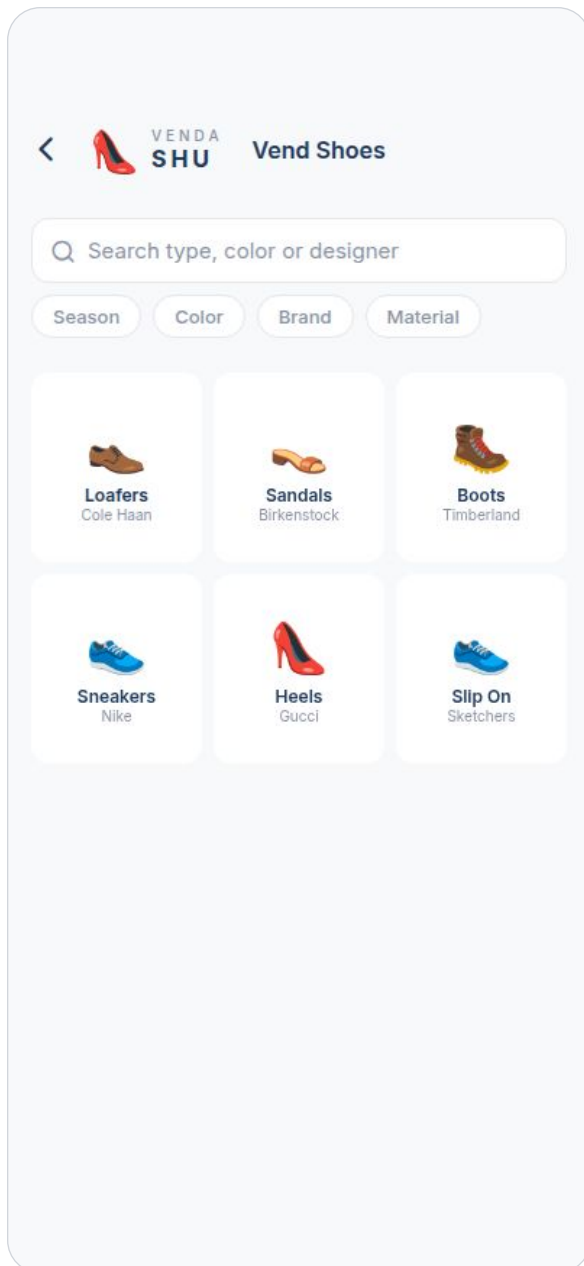


Botanical

Blurred greenery on a cream wall; fresh and natural.

After the photo, step two of intake collects the details: shoe type, construction/material, season, color, designer, and subsection. The system then assigns the pair to an appropriate bin automatically and shows you the final record (photo included) for confirmation before saving.

Vend Shoes



The Vend screen is your full catalog: every pair currently stored in the unit, each showing its studio photo (or a shoe-type icon if no photo was taken) with type and designer beneath.

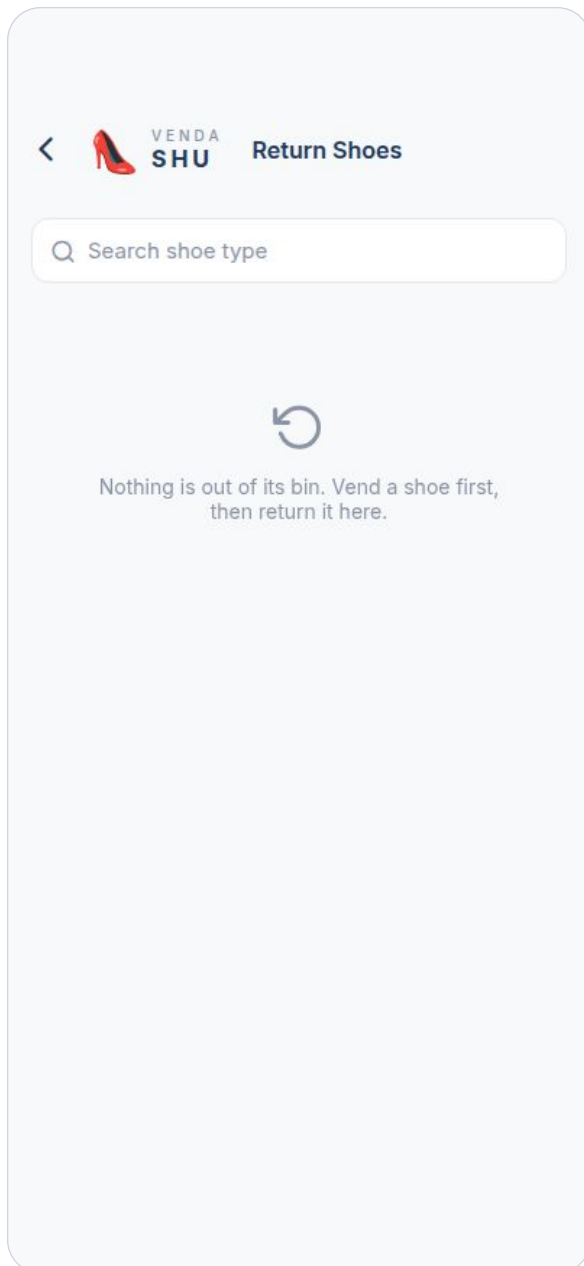
Finding and vending a pair

- Search bar — matches against type, color, and designer as you type.
- Filter chips — Season, Color, Brand, and Material each open a filter; combine them to narrow a large collection quickly.
- Tap a pair — opens a detail view with the full photo and every recorded attribute, plus the exact bin location (column, row, LF/RF).
- Confirm vend — the unit ejects the bin holding your shoes and lights its LED so you can find it at a glance. The pair's status changes to “out,” the Home counters update, and it appears on the Return screen.

Note for the meticulous:

Only pairs with status “stored” appear here — a pair that is already out cannot be vended a second time; the system enforces this at the database level, not just in the interface.

Return Shoes



Everything currently out of the unit is listed here. The screenshot shows the empty state — the message confirms nothing is out, which is exactly what you want to see at the end of the day.

Returning a pair

- Search bar — filter by shoe type if several pairs are out at once.
- Tap the pair you are physically holding — a confirmation view shows its photo and its assigned bin.
- Confirm — the unit opens that pair's own bin (never a different one), you place the shoes inside, and the status flips back to “stored.”
- The Home screen's “Out of bins” counter decreases immediately.

Note for the meticulous:

Because every pair has a permanent assigned bin, shoes always return to the same location — the system never shuffles pairs between bins, so physical organization stays consistent forever.

Label Printing with the ColAura

Every pair stored in Vend-a-Shu carries label text in its record — a compact summary of the shoe's type, owner, and bin assignment. These print as full-color physical labels using the Brother ColAura (VC-500W) color photo & label printer, so each bin can be labeled with the very photo and details stored in the app.

About the printer

- Full-color, ink-free printing (ZINK® zero-ink technology) — no cartridges to replace.
- Prints labels up to 2 3 wide and 17 3 long at 313 dpi — room for a shoe photo, owner name, and bin coordinates on one label.
- Connects over Wi-Fi, so the VAS system can send labels directly without cables.
- Compact enough to store in the top drawer of the VAS unit — that is where yours is kept.



Brother ColAura (VC-500W) color photo & label printer

What labels are for

- Labels combine the shoe details stored in the app with the bin coordinates (column / row / location), so a bin can be identified at a glance without opening the app.
- Every pair's label text is already recorded in the system at intake.

Documentation status

Detailed documentation for the ColAura label printing workflow is yet to be added to this guide. It will be included in a future revision of this document.

SECTION 11

Troubleshooting & FAQ

Q. The photo background removal fails with an error.

A. Retry with a clearer photo: good lighting, shoes fully in frame, JPEG or PNG format. The app reports the exact reason (invalid image vs. processing failure). Processing takes ~5 seconds — the spinner is normal, not a hang.

Q. I don't see the newest features in my app.

A. The published (live) app updates only when a new version is published. If a feature exists in preview but not in your app, a republish is needed.

Q. The backgrounds row is empty.

A. The backdrop images live on the VAS server. Verify the app is connected (Connect screen shows the green online badge) and try again.

Q. A pair shows an icon instead of a photo.

A. That pair was added with "Continue without photo." Photos are attached at intake; to add one, remove and re-add the pair with a photo.

Q. Can two people use the same unit?

A. Yes. Create a user per person in Options; shoes are recorded per owner. Use Switch User on the Home screen to change the active person, and turn off "Remember me" on shared devices.

Q. What are the photo limits?

A. JPEG or PNG input; the processed image is downscaled to 900 px on its longest side and must be under 3 MB — both are handled automatically, so ordinary phone photos always fit.

Q. The unit ejected the wrong bin / nothing happened.

A. Check the bin coordinates shown on the confirmation screen against the unit's labels. If the mismatch persists, power-cycle the unit and reconnect from the Connect screen.

Q. Can I access my shoe catalog away from home?

A. Yes — browsing, searching, and reviewing your collection works remotely, e.g. while shopping, so you can check what you already own before buying. Only vending and returning require you to be physically at the unit.

Q. Can I print labels for the shoe bins?

A. Yes — shoe labels print via the CoLAura printer stored in the top drawer of the unit. Detailed documentation for the label printing workflow is yet to be added to this guide (see Section 10).

Prepared by Brainchild Engineering

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